

Text Mining for Political & Social Science

Course syllabus. Instructor: Kirill Maslinsky.

IPSA Summer School on Methods in Political Science, 2018. MA and PhD level, 14 hours.

A two-part summer school workshop. Part I is an introductory to intermediate level course on words as units of operationalization and analysis; Part II is an intermediate to advanced level course on modelling text semantics as space and tracing change of meaning in time. The parts share their motivation, toolbox and audience, and may be taken separately or in sequence.

Instructor

Kirill Maslinsky, Higher School of Economics, Candidate of Sciences in Anthropology.

Kirill Maslinsky is a research fellow at the Laboratory of Sociology of Education and Science and lecturer in the Department of Sociology, Higher School of Economics — Saint Petersburg. He was trained as a linguist and an anthropologist. Later he became interested in the application of methods of computational linguistics to various problems and data in the social sciences and humanities, which is now the topic of his teaching and research. His research experience includes both quantitative and qualitative methods. He also authored a textbook on the Linux operating system, and was instrumental in building the software infrastructure for linguistic corpora in several African languages.

Course Overview

For the social and political sciences, written texts provide essential data for studying ideology and political discourse, conflict, sentiment and political affiliation, among many other things. With a growing availability of larger collections of text in digital form it is tempting to scale the research up in terms of the population studied (e.g. “all of twitter”), time spans (e.g. “all of the American history”), and geographical scope (e.g. “all foreign ties of China”). Computational methods for text analysis promise to aid at the scale where traditional content analysis is not feasible.

The goal of the workshop is to provide a basic understanding of how to properly use collections of texts as quantitative evidence, and to make this knowledge practical. We will use the R programming environment as a toolbox for text analysis. To “learn by doing” we will work with real text collections and will replicate some methods from recent social research employing computational text analysis.

Target Group / Suggested Level of Study

Master and PhD students in political science, international relations, sociology.

Prerequisites

Students are assumed to be familiar with traditional content analysis, and to have basic knowledge of statistics. Some background knowledge in linguistics will be helpful, but is not critical. The R programming environment will be used for practical exercises, so basic knowledge of R is desirable. Reading any introductory book or tutorial on R, or taking the preparatory “Introduction to Statistics in R” course during the Summer school, are good options for students without prior knowledge of R.

Part I. Counting Words

The focus of this introductory to intermediate level course is how words can be used as units of operationalization and analysis. Essentially, counting words is the backbone of many baseline and more complicated statistical models of text. We will discuss the impact of preprocessing text, word frequency distributions, word co-occurrence patterns, and vector representation of text. The methods considered in this course include dictionary-based methods, analysis of the word co-occurrence graph, supervised machine learning (document classification) and unsupervised topic modeling (LDA). Validation of the methods and the results will be an important cross-cutting theme.

Indicative Content / Keywords

Natural language processing, Zipf's law, distributional semantics, supervised machine learning, document classification, topic modeling, LDA; R packages: quanteda, tidytext, tm, caret, mallem.

Intended Learning Outcomes

- Understanding the possibilities of automated text analysis as well as its pitfalls, and important caveats about applying statistical tests to language data.
- Being able to apply computational methods of text analysis (e.g. analysis of word frequency and co-occurrence, document classification, topic modeling) to collections of texts.
- Being able to adequately interpret and report the results of computational text analysis in research papers.

Essential Readings

- Aggarwal, C. C., & Zhai, C. (Eds.). (2012). Mining Text Data. Springer Science & Business Media.
- Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3), 267–297.
- Rule, A., Cointet, J. P., & Bearman, P. S. (2015). Lexical shifts, substantive changes, and continuity in State of the Union discourse, 1790–2014. *Proceedings of the National Academy of Sciences*, 112(35), 10837–10844.

Part II. Semantic Spaces

The focus of this intermediate to advanced level course is on modelling text semantics via a spatial metaphor and on modelling change of meaning in time. The spatial metaphor underlies all models where the meanings of words and texts are represented as points in a multidimensional space. In political science, spatial models are commonly used to represent ideological scales, political affiliation and sentiment. During the course we will encounter several unsupervised and weakly supervised methods for constructing such semantic spaces from collections of texts, and will learn how to structure and navigate these spaces. We will discuss document clustering (including topic modeling), word embeddings (word2vec) and other methods of dimensionality reduction. Using spatial models to trace change of meaning and of ideological landscapes in time will be an important cross-cutting theme.

Indicative Content / Keywords

Natural language processing, distributional semantics, dimensionality reduction, unsupervised machine learning, document clustering, sentiment analysis, topic modeling; R packages: quanteda, tidytext, mallet, wordVectors.

Intended Learning Outcomes

- Understanding multidimensional representation of lexical meaning and the role of dimensionality reduction.
- Being able to apply word embedding and clustering methods to downstream tasks, such as sentiment analysis, ideological scaling etc.
- Being able to adequately interpret spatial models representing meaning and to report the results of research employing these models.

Essential Readings

- Jurafsky, D., & Martin, J. H. (2008). *Speech and Language Processing*, 2nd edition. NJ: Prentice Hall. (And some chapters from the 3rd edition draft, available online: <http://web.stanford.edu/~jurafsky/slp3/>)
- Pozzi, F. A., Fersini, E., Messina, E., & Liu, B. (2016). *Sentiment Analysis in Social Networks*. Morgan Kaufmann.

Credits

ECTS 2 per part.